

CBSE CLASS 10 MATHEMATICS

Formula & Concept Bible

2025–26 Board Exam Edition · Official CBSE Syllabus

All 7 Units · 150+ Formulas · Key Theorems · Worked Examples · PYQs

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Official Syllabus Overview 2025-26 (Theory = 80 Marks)

Unit	Chapters	Marks
I: Number Systems	Ch 1: Real Numbers	06
II: Algebra	Ch 2: Polynomials Ch 3: Linear Eqns Ch 4: Quadratic Ch 5: AP	20
III: Coord. Geometry	Ch 7: Distance & Section Formula	06
IV: Geometry	Ch 6: Triangles (Similarity) Ch 10: Circles (Tangents)	15
V: Trigonometry	Ch 8: Intro to Trig Ch 9: Heights & Distances	12
VI: Mensuration	Ch 11: Areas Related to Circles Ch 12: SA & Volumes	10
VII: Stats & Prob.	Ch 13: Statistics Ch 14: Probability	11
	TOTAL THEORY	80

**How to Use This Book**

- Every chapter: Concept Summary → Graphs → Key Theorems → OMML Formulas → Common Mistakes → Worked Example → PYQs
- 📌 Important boxes = high-probability exam topics
- ⚠ Mistake boxes = traps that cost marks every year
- 🧠 Concept boxes = step-by-step method/proof outlines
- 💡 Tip boxes = shortcut techniques and exam tricks
- Visit sjmaths.com for free video solutions to every PYQ listed here



CHAPTER 1 · Unit I — 6 Marks

Real Numbers

Fundamental Theorem of Arithmetic | Irrationality Proofs | HCF & LCM

1.1 Key Concepts

- ▶ Natural Numbers $\subset$ Whole Numbers $\subset$ Integers $\subset$ Rational Numbers $\subset$ Real Numbers
- ▶ Rational numbers $\leftrightarrow$ terminating or recurring decimals (e.g. $3/4 = 0.75$, $1/3 = 0.333\dots$)
- ▶ Irrational numbers $\leftrightarrow$ non-terminating, non-recurring decimals (e.g. $\sqrt{2}$, π , e)

1.2 Fundamental Theorem of Arithmetic

Fundamental Theorem

- Every composite number can be expressed as a product of primes in a UNIQUE way (order apart).
- $\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$ (valid for TWO numbers only)
- HCF $\rightarrow$ take MINIMUM powers of COMMON prime factors
- LCM $\rightarrow$ take MAXIMUM powers of ALL prime factors

▶ $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

Example: HCF and LCM of 12 and 18

- ▶ $12 = 2^2 \times 3^1$ $18 = 2^1 \times 3^2$
- ▶ $\text{HCF} = 2^1 \times 3^1 = 6$ (minimum powers of common factors: 2 and 3)
- ▶ $\text{LCM} = 2^2 \times 3^2 = 36$ (maximum powers of all factors: 2 and 3)
- ▶ Check: $\text{HCF} \times \text{LCM} = 6 \times 36 = 216 = 12 \times 18$ ✓

1.3 Irrationality Proofs (Proof by Contradiction)

Proof that $\sqrt{2}$ is Irrational

- Assume $\sqrt{2} = p/q$ where $\text{HCF}(p, q) = 1$ (rational in lowest terms)
- Then $2 = p^2/q^2 \Rightarrow p^2 = 2q^2 \Rightarrow 2$ divides $p^2 \Rightarrow 2$ divides p
- Let $p = 2m$. Then $4m^2 = 2q^2 \Rightarrow q^2 = 2m^2 \Rightarrow 2$ divides q
- But $\text{HCF}(p, q) = 1$. CONTRADICTION. $\therefore \sqrt{2}$ is irrational.

Irrationality Rules (Most Tested)

- $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$ are all irrational
- Rational + irrational = irrational (e.g. $3 + \sqrt{2}$)
- Non-zero rational $\times$ irrational = irrational (e.g. $2\sqrt{3}$)
- Prove $3 + 2\sqrt{5}$: assume rational $\Rightarrow \sqrt{5} = (\text{rational}-3)/2 = \text{rational}$ — contradiction!

⚠ Common Mistakes

- $\text{HCF} \times \text{LCM} = a \times b$ works for TWO numbers only, NOT three.
- Irrationality proof: must state $\text{HCF}(p,q) = 1$ at the start.
- Terminating decimal $\leftrightarrow$ denominator (in lowest form) has ONLY $2^s \times 5^t$ as factors.

1.4 Worked Example**➡ Worked Example**

Q: Find HCF and LCM of 96 and 120. Verify $\text{HCF} \times \text{LCM} = \text{product}$.

Step 1: $96 = 2^5 \times 3$ $120 = 2^3 \times 3 \times 5$

Step 2: $\text{HCF} = 2^3 \times 3 = 24$ (min powers of common factors 2, 3)

Step 3: $\text{LCM} = 2^5 \times 3 \times 5 = 480$ (max powers of all factors)

Step 4: Verify: $24 \times 480 = 11520 = 96 \times 120$ ✓

∴ Answer: $\text{HCF} = 24$, $\text{LCM} = 480$

1.5 PYQ Triggers

Year	PYQ — Board Exam Question
2024	Show that $3 + 2\sqrt{5}$ is irrational, given $\sqrt{5}$ is irrational. [2 marks]
2023	Find LCM and HCF of 867 and 255. Verify $\text{HCF} \times \text{LCM} = \text{product}$. [3 marks]
2022	Prove that $\sqrt{3}$ is irrational. [2 marks]
2021	Given $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$. [1 mark]
2019	Show that $5 - \sqrt{3}$ is irrational. [2 marks]

CHAPTER 2 · Unit II — Part of 20 Marks

Polynomials

Zeros & Coefficients Relationship | Quadratic | Graphical Interpretation

2.1 Types of Polynomials

Type	General Form	Degree	Graph Shape	Max Zeros
Linear	$ax + b, a \neq 0$	1	Straight line	1
Quadratic	$ax^2 + bx + c, a \neq 0$	2	Parabola (U or $\cap$)	2
Cubic	$ax^3 + bx^2 + cx + d$	3	S-curve	3

2.2 Zeros and Coefficients of a Quadratic

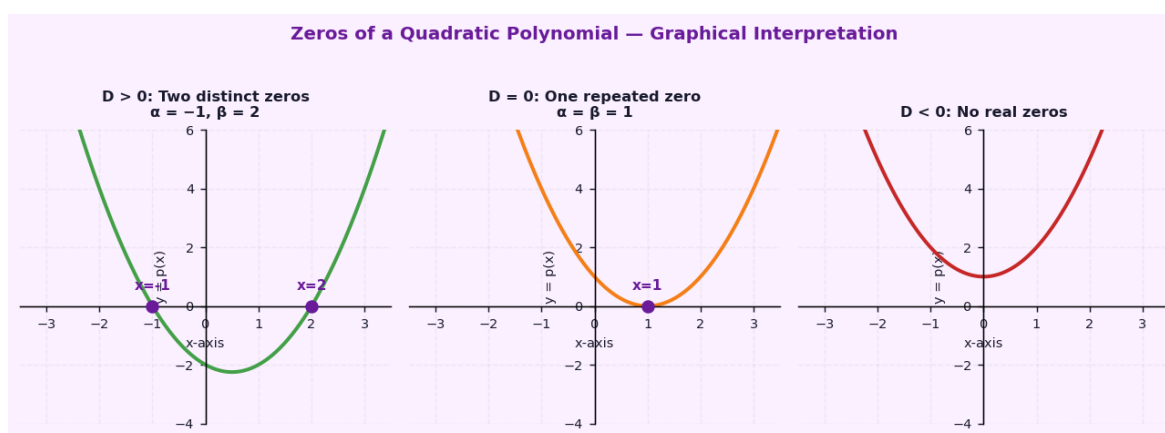
🔑 Key Relationship (α, β are zeros of $ax^2 + bx + c = 0$)

- Sum of zeros: $\alpha + \beta = -b/a$
- Product of zeros: $\alpha\beta = c/a$
- Quadratic polynomial from zeros: $x^2 - (\alpha + \beta)x + \alpha\beta$

- Sum of zeros: $\alpha + \beta = \frac{-b}{a}$
- Product of zeros: $\alpha\beta = \frac{c}{a}$
- Polynomial = $x^2 - (\alpha + \beta)x + \alpha\beta$

2.3 Graphical Interpretation of Zeros

- Zeros = x-coordinates where the graph CROSSES or TOUCHES the x-axis
- $D > 0$: parabola cuts x-axis at 2 points → two distinct real zeros
- $D = 0$: parabola TOUCHES x-axis at 1 point → one repeated zero
- $D < 0$: parabola does NOT cut x-axis → no real zeros



⚠ Common Mistakes

- Sum = $-b/a$ (NEGATIVE b over a). Students often write $+b/a$.
- Product = c/a (positive c over a). Same signs as in the original polynomial.
- Zero of polynomial $\neq$ value of polynomial. Zero is where $p(x) = 0$.
- If graph TOUCHES x-axis (doesn't cross): repeated zero $\alpha = \beta$, so $D = 0$.

2.4 Worked Example**➡ Worked Example**

Q: Find a quadratic polynomial whose zeros are 3 and -4.

Step 1: Sum = $3 + (-4) = -1$

Step 2: Product = $3 \times (-4) = -12$

Step 3: Polynomial = $x^2 - (\text{sum})x + \text{product} = x^2 - (-1)x + (-12)$

∴ Answer: $x^2 + x - 12$

2.5 PYQ Triggers

Year	PYQ — Board Exam Question
2024	If sum of zeros of $kx^2+2x+3k$ equals their product, find k. [2 marks]
2023	Find zeros of $2x^2-7x-13$ and verify sum/product relations. [3 marks]
2022	If α, β are zeros of $4x^2-5x+2$, find $1/\alpha + 1/\beta$. [2 marks]
2021	Find a quadratic polynomial with sum of zeros -3 and product 2. [2 marks]
2019	Find the zeros of $4\sqrt{3}x^2+5x-2\sqrt{3}$. [2 marks]

CHAPTER 3 · Unit II — Part of 20 Marks

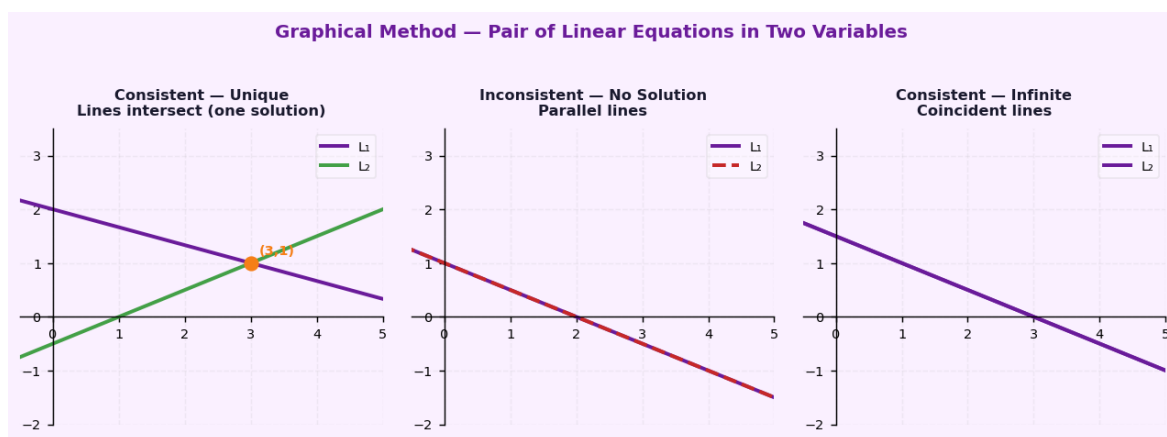
Pair of Linear Equations in Two Variables

Graphical Method | Substitution | Elimination | Consistency

3.1 General Form & Consistency Conditions

► Standard form: $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$

Condition	Type	Graph	Solutions
$a_1/a_2 \neq b_1/b_2$	Consistent — Unique	Lines intersect	Exactly one
$a_1/a_2 = b_1/b_2 = c_1/c_2$	Consistent — Infinite	Lines coincide	Infinitely many
$a_1/a_2 = b_1/b_2 \neq c_1/c_2$	Inconsistent	Lines are parallel	No solution



Graph 3.1: Three cases for a pair of linear equations — intersecting, parallel, coincident

3.2 Methods of Solution

Substitution Method



Steps

- Step 1: Express one variable (say y) in terms of x from one equation.
- Step 2: Substitute into the second equation → one equation in one variable.
- Step 3: Solve. Back-substitute to find the other variable.

Elimination Method



Steps

- Step 1: Multiply equations to make the coefficient of one variable equal.
- Step 2: Add or subtract to eliminate that variable.
- Step 3: Solve the resulting equation. Back-substitute.



⚠ Common Mistakes

- Always check the consistency condition BEFORE solving — saves time.
- For word problems: define variables clearly at the start.
- When lines are parallel: the system has NO solution (not ∞ solutions).

3.3 Worked Example**➡ Worked Example**

Q: Solve: $2x + 3y = 13$ and $5x - 4y = -2$

Step 1: Elimination: multiply eq1 by 4, eq2 by 3:

Step 2: $8x + 12y = 52$ and $15x - 12y = -6$

Step 3: Add: $23x = 46 \Rightarrow x = 2$

Step 4: Substitute $x = 2$ in eq1: $4 + 3y = 13 \Rightarrow y = 3$

$\therefore$ Answer: $x = 2, y = 3$

3.4 PYQ Triggers

Year	PYQ — Board Exam Question
2024	Solve graphically: $x+3y=6$ and $2x-3y=12$. Find area of triangle formed with axes. [4 marks]
2023	Boat: 18 km upstream + 24 km downstream in 6 hr; 12 km up + 36 km down in 6 hr. Find speeds. [4 marks]
2022	For what value of k will $3x+y=1$ and $(2k-1)x+(k-1)y=2k+1$ have no solution? [2 marks]
2021	Find a, b for which $2x+3y=7$ and $(a+b)x+(2a-b)y=21$ has infinitely many solutions. [3 marks]
2019	Solve: $(a-b)x+(a+b)y=a^2-2ab-b^2$ and $(a+b)(x+y)=a^2+b^2$. [3 marks]

CHAPTER 4 · Unit II — Part of 20 Marks

Quadratic Equations

Standard Form | Factorisation | Quadratic Formula | Discriminant

4.1 Standard Form & Methods

► Standard form: $ax^2 + bx + c = 0$, where $a \neq 0$

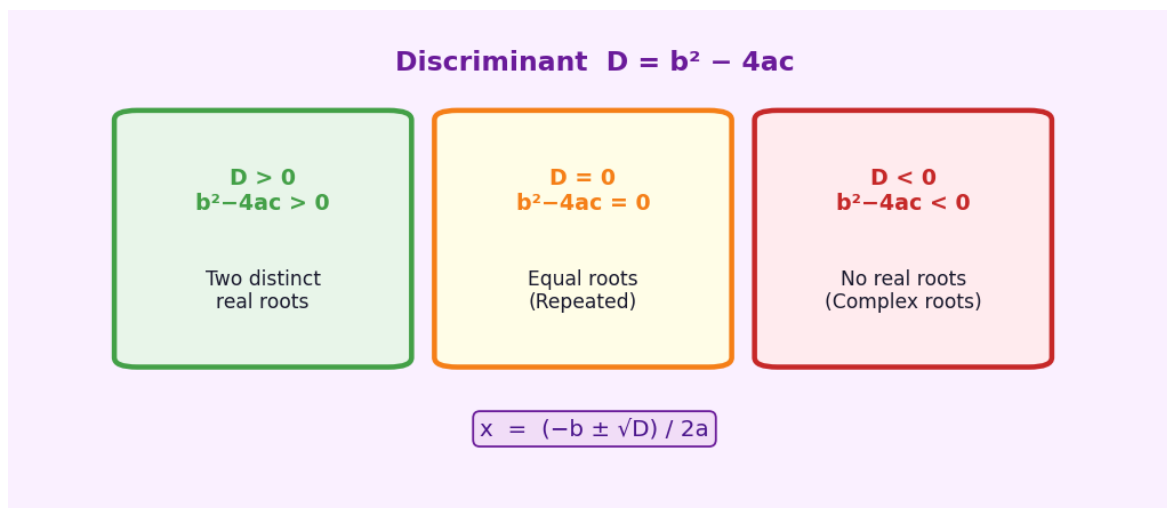
Method 1 — Factorisation (split the middle term)

- Find two numbers whose sum = b and product = ac
- Example: $x^2 + 5x + 6 = 0 \Rightarrow (x+2)(x+3) = 0 \Rightarrow x = -2$ or $x = -3$

Method 2 — Quadratic Formula

► **Quadratic Formula:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ *Use when factorisation is not straightforward*

4.2 Discriminant — Nature of Roots

► **Discriminant:** $D = b^2 - 4ac$ Graph 4.1: Nature of roots based on value of discriminant D

Value of D	Nature of Roots	Example
$D > 0$	Two distinct real roots	$x^2 - 5x + 6 = 0$: $D = 1 > 0 \rightarrow$ roots 2, 3
$D = 0$	Two equal (repeated) real roots	$x^2 - 4x + 4 = 0$: $D = 0 \rightarrow$ root 2, 2
$D < 0$	No real roots	$x^2 + x + 1 = 0$: $D = -3 < 0$

Key Facts

- Sum of roots: $\alpha + \beta = -b/a$ | Product of roots: $\alpha\beta = c/a$
- For equal roots: set $D = 0$, i.e. $b^2 = 4ac \rightarrow$ find unknown coefficient
- For real roots: set $D \geq 0$

⚠ Common Mistakes

- Formula denominator is $2a$, NOT just 2. Write $x = (-b \pm \sqrt{D}) / (2a)$.
- $D = b^2 - 4ac$. Students often write $4ac - b^2$ (wrong sign).
- $D = 0$ means EQUAL roots, NOT no roots.

4.3 Worked Example

➡ Worked Example

Q: Find the value of k so that $x^2 - 4x + k = 0$ has equal roots.

Step 1: For equal roots: $D = 0 \Rightarrow b^2 - 4ac = 0$

Step 2: Here $a = 1$, $b = -4$, $c = k$

Step 3: $(-4)^2 - 4(1)(k) = 0 \Rightarrow 16 - 4k = 0 \Rightarrow k = 4$

∴ Answer: $k = 4$

4.4 PYQ Triggers

Year	PYQ — Board Exam Question
2024	Find the discriminant of $x^2 + 5x + 5 = 0$ and state the nature of roots. [1 mark]
2023	Train travels 480 km. If speed were 8 km/h more, time decreases by 2 hr. Find speed. [4 marks]
2022	Find k so that $(k-12)x^2 + 2(k-12)x + 2 = 0$ has equal roots. [2 marks]
2021	Solve: $4x^2 - 4ax + (a^2 - b^2) = 0$. [3 marks]
2019	Sum of areas of two squares is 468 m^2 . Difference of perimeters is 24 m. Find sides. [4 marks]

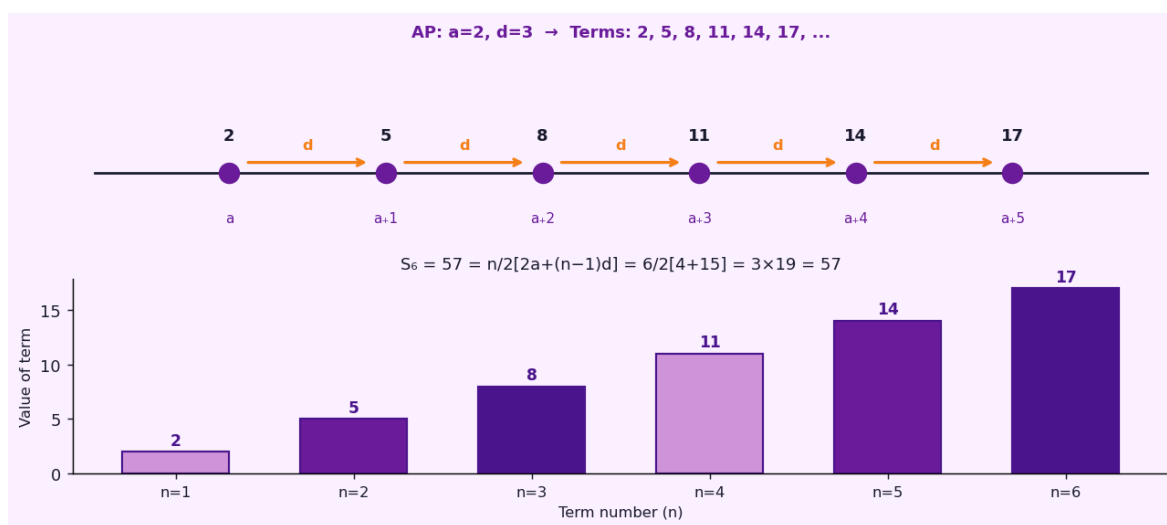
CHAPTER 5 · Unit II — Part of 20 Marks

Arithmetic Progressions (AP)

nth term | Sum of n terms | Applications

5.1 Definition & Key Terms

- ▶ AP: a sequence where consecutive terms differ by a **CONSTANT** amount (common difference d)
- ▶ General form: $a, a+d, a+2d, a+3d, \dots$
- ▶ a = first term | d = common difference | n = number of terms | l = last term
- ▶ $d = a_2 - a_1 = a_3 - a_2$ (must be the same for every pair of consecutive terms)



Graph 5.1: AP as a number line (top) and bar chart of terms (bottom)

5.2 Core Formulas

- ▶ **n^{th} term:** $a_n = a + (n - 1)d$ ★ *Most used formula in board exams*
- ▶ **Sum of first n terms:** $S_n = \frac{n}{2} [2a + (n - 1)d]$
- ▶ **Also (when last term l is known):** $S_n = \frac{n}{2} (a + l)$

Important Results

- Three terms in AP: use $(a-d), a, (a+d) \rightarrow \text{sum} = 3a$ (very useful in problems)
- Four terms in AP: use $(a-3d), (a-d), (a+d), (a+3d) \rightarrow \text{sum} = 4a$
- If S_n is given, find a_n using: $a_n = S_n - S_{n-1}$
- If $a_m = n$ and $a_n = m$, then $a_{m+n} = 0$

Common Mistakes

- $a_n = a + (n-1)d$: coefficient is $(n-1)$, NOT n .
- S_n formula: $2a + (n-1)d$ inside brackets, not $2a + nd$.

- If S_n given as pn^2+qn : use $a_n = S_n - S_{n-1}$ (don't use $a_n=a+(n-1)d$ directly).
- d can be negative (decreasing AP) or zero (constant sequence).

5.3 Worked Example

Worked Example

Q: $S_{20} = 400$ and $S_{40} = 1600$ for an AP. Find S_{60} .

Step 1: $S_{20} = 20/2 [2a+19d] = 400 \Rightarrow 2a+19d = 40 \dots(i)$

Step 2: $S_{40} = 40/2 [2a+39d] = 1600 \Rightarrow 2a+39d = 80 \dots(ii)$

Step 3: Subtract (i) from (ii): $20d = 40 \Rightarrow d = 2$. From (i): $2a = 2 \Rightarrow a = 1$

Step 4: $S_{60} = 60/2 [2(1)+59(2)] = 30[2+118] = 30 \times 120$

∴ Answer: $S_{60} = 3600$

5.4 PYQ Triggers

Year	PYQ — Board Exam Question
2024	First term of AP is -5 , common difference 2 . Find its 12th term. [1 mark]
2023	How many terms of AP $9, 17, 25, \dots$ must be taken to give sum 636 ? [3 marks]
2022	9th term of AP is 0 . Prove 29th term = twice the 19th term. [2 marks]
2021	Find sum of all 3-digit natural numbers divisible by 5 . [3 marks]
2019	Ratio of 11th to 18th term of AP is $2:3$. Find ratio of 5th to 21st term. [3 marks]

CHAPTER 6 · Unit IV — Part of 15 Marks

Triangles

Similarity | BPT | AA / SSS / SAS Criteria | Pythagoras Theorem

6.1 Similarity vs Congruence

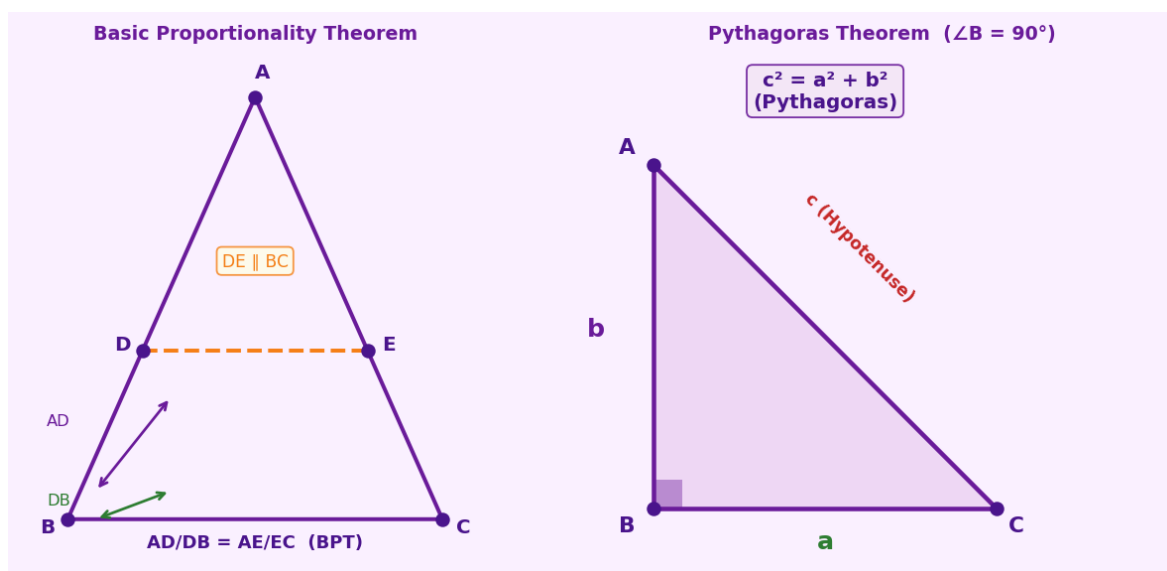
	Similarity	Congruence
Symbol	$\sim$	$\cong$
Shape	Same shape	Same shape AND size
Angles	All equal	All equal
Sides	Corresponding sides proportional	Corresponding sides equal

6.2 Basic Proportionality Theorem (BPT) ★

BPT / Thales' Theorem (PROOF required in exams)

- If $DE \parallel BC$ in $\triangle ABC$, then: $AD/DB = AE/EC$
- Converse: If $AD/DB = AE/EC$, then $DE \parallel BC$
- Proof: draw $DM \perp AC$, $EN \perp AB$. Use equal-area argument to show ratios equal.

► BPT: $\frac{AD}{DB} = \frac{AE}{EC}$



Graph 6.1: BPT (left) and Pythagoras Theorem (right)

6.3 Criteria for Similarity of Triangles

Criterion	Condition	Important Note
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AA	Two angles of $\triangle ABC$ = two angles of $\triangle DEF$	Most commonly used
SSS	$AB/DE = BC/EF = CA/FD$ (all three sides proportional)	Check all three ratios
SAS	One angle equal + two sides about that angle proportional	Angle must be INCLUDED

6.4 Area Theorem

Ratio of Areas of Similar Triangles

- Ratio of areas = square of ratio of corresponding sides
- $\text{ar}(\triangle ABC) / \text{ar}(\triangle DEF) = (AB/DE)^2 = (BC/EF)^2 = (CA/FD)^2$

► $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \left(\frac{AB}{DE}\right)^2$

6.5 Pythagoras Theorem (PROOF required)

Pythagoras Theorem

- In a right triangle, $(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$
- $c^2 = a^2 + b^2$ where c is the hypotenuse (side opposite 90°)
- Converse: if $c^2 = a^2 + b^2$, then the angle opposite c is 90°

► **Pythagoras:** $c^2 = a^2 + b^2$ $c = \text{hypotenuse (opposite } 90^\circ)$

Common Mistakes

- BPT: DE must be PARALLEL to BC — check this condition first.
- SAS: the equal angle must be BETWEEN the proportional sides (included).
- Area ratio = side ratio SQUARED. (Volumes use cube, not square.)
- Always identify hypotenuse correctly — it is the side opposite the right angle.

6.6 Worked Example

→ Worked Example

Q: $\triangle ABC \sim \triangle DEF$. $AB = 4$ cm, $DE = 6$ cm, $\text{ar}(\triangle ABC) = 16$ cm². Find $\text{ar}(\triangle DEF)$.

Step 1: $\text{ar}(\triangle ABC)/\text{ar}(\triangle DEF) = (AB/DE)^2 = (4/6)^2 = 16/36$

Step 2: $16 / \text{ar}(\triangle DEF) = 16/36$

Step 3: $\text{ar}(\triangle DEF) = 36$ cm²

∴ **Answer:** $\text{ar}(\triangle DEF) = 36$ cm²

6.7 PYQ Triggers

Year	PYQ — Board Exam Question
2024	In $\triangle ABC$, D on AB, E on AC, $DE \parallel BC$. $AD=6$, $DB=9$, $AE=8$. Find EC. [2 marks]

2023	State and prove Basic Proportionality Theorem. [5 marks]
2022	$\triangle ABC \sim \triangle QRP$. $\text{ar}(\triangle ABC)/\text{ar}(\triangle QRP)=9/4$, $BC=15$ cm. Find PR . [2 marks]
2021	Prove Pythagoras Theorem. Hence find BC if $\angle B=90^\circ$, $AB=5$, $AC=13$. [5 marks]
2019	In $\triangle ABC$, $\angle B=90^\circ$. D and E on BC and AC . Prove $\triangle ABD \sim \triangle ECB$. [3 marks]

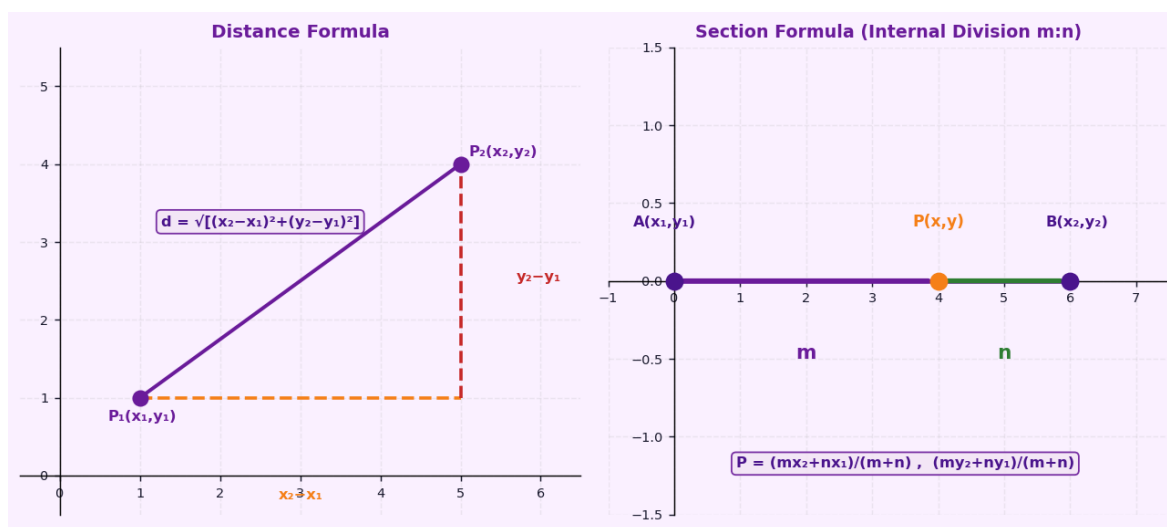
CHAPTER 7 · Unit III — 6 Marks

Coordinate Geometry

Distance Formula | Section Formula | Midpoint Formula

7.1 Distance Formula

- Distance between $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Distance from origin $O(0,0)$ to $P(x, y)$: $d = \sqrt{x^2 + y^2}$



Graph 7.1: Distance formula (left) and Section formula (right)

Uses of Distance Formula

- Equilateral $\triangle$: all 3 sides equal | Isosceles $\triangle$: any 2 sides equal | Scalene $\triangle$: no sides equal
- Right-angled $\triangle$: check $a^2 + b^2 = c^2$ (Pythagoras)
- Collinear points: $AB + BC = AC$ (total distance equals sum of parts)
- Type of quadrilateral: find all 4 sides AND 2 diagonals

7.2 Section Formula

- Internal division in ratio $m : n$: $P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
- Midpoint ($m = n = 1$): $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
- Area of $\triangle$ (three vertices): $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
Take absolute value!

Common Mistakes

- Section formula: m goes with (x_2, y_2) , NOT (x_1, y_1) . m is the ratio closer to P_2 .

- Midpoint: add then divide by 2 (not subtract).
- Area of triangle: result must be positive — take the absolute value!
- Collinearity check: area of triangle = 0 OR slopes of AB = BC.

7.3 Worked Example

Worked Example

Q: Find the point dividing the line joining (4, -3) and (8, 5) in ratio 3:1 internally.

Step 1: $m = 3, n = 1, (x_1, y_1) = (4, -3), (x_2, y_2) = (8, 5)$

Step 2: $x = (3 \times 8 + 1 \times 4) / (3 + 1) = (24 + 4) / 4 = 7$

Step 3: $y = (3 \times 5 + 1 \times (-3)) / (3 + 1) = (15 - 3) / 4 = 3$

∴ Answer: Point = (7, 3)

7.4 PYQ Triggers

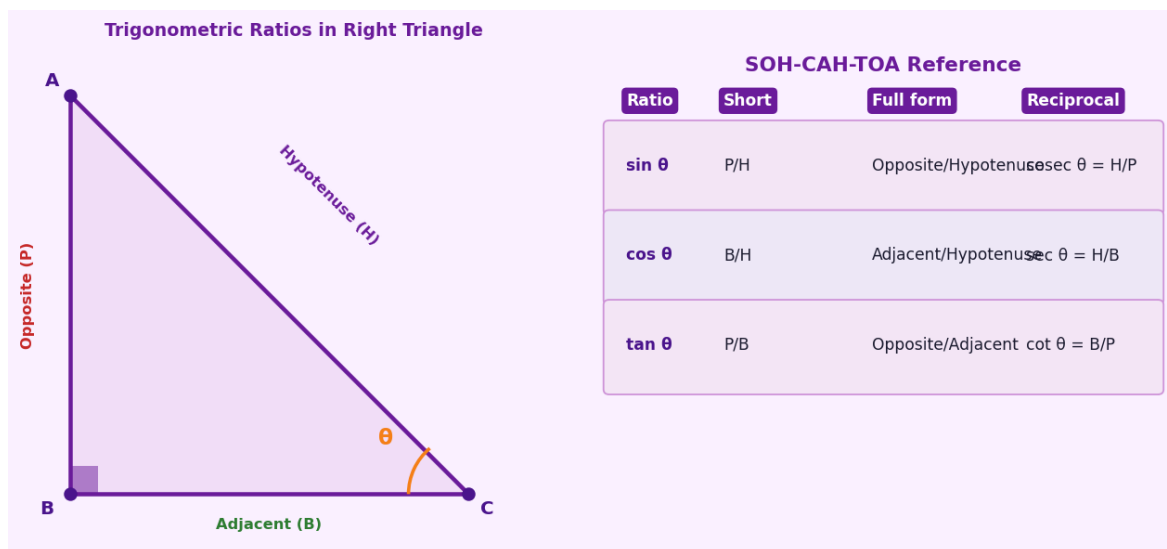
Year	PYQ — Board Exam Question
2024	Find the ratio in which $(-3, k)$ divides the join of $(-5, -4)$ and $(-2, 3)$. Find k . [3 marks]
2023	Show that $A(6,1), B(8,2), C(9,4), D(7,3)$ form a parallelogram. [3 marks]
2022	Find the centre of the circle passing through $(6,6), (3,7), (3,3)$. [3 marks]
2021	If $(1,2), (4,y), (x,6), (3,5)$ are vertices of a parallelogram, find x and y . [2 marks]
2019	Find area of triangle formed by midpoints of a $\triangle$ with vertices $(0,-1), (2,1), (0,3)$. [3 marks]

CHAPTER 8 · Unit V — Part of 12 Marks

Introduction to Trigonometry

Trig Ratios | Standard Values | Identities | Complementary Angles

8.1 Trigonometric Ratios



Graph 8.1: Right triangle with labelled sides (left) and SOH-CAH-TOA reference (right)

Ratio	Definition	Reciprocal
$\sin \theta$	Opposite / Hypotenuse (P/H)	$\operatorname{cosec} \theta = 1/\sin \theta$
$\cos \theta$	Adjacent / Hypotenuse (B/H)	$\sec \theta = 1/\cos \theta$
$\tan \theta$	Opposite / Adjacent = $\sin \theta / \cos \theta$ (P/B)	$\cot \theta = 1/\tan \theta$

8.2 Standard Values Table

Ratio	0°	30°	45°	60°	90°
$\sin \theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	Undef.
$\operatorname{cosec} \theta$	Undef.	2	$\sqrt{2}$	$2/\sqrt{3}$	1
$\sec \theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	Undef.
$\cot \theta$	Undef.	$\sqrt{3}$	1	$1/\sqrt{3}$	0

💡 Trick to Remember sin Values (0° to 90°)

- Write: $\sqrt{0/2}$, $\sqrt{1/2}$, $\sqrt{2/2}$, $\sqrt{3/2}$, $\sqrt{4/2}$
- = 0, $1/2$, $1/\sqrt{2}$, $\sqrt{3}/2$, 1 (sin 0° , 30° , 45° , 60° , 90°)
- cos is exactly the REVERSE order of sin.

8.3 Trigonometric Identities

- ▶ $\sin^2\theta + \cos^2\theta = 1$ ← *Foundation identity. Memorise first!*
- ▶ $1 + \tan^2\theta = \sec^2\theta$ *Derived: divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$*
- ▶ $1 + \cot^2\theta = \operatorname{cosec}^2\theta$ *Derived: divide $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$*

8.4 Complementary Angle Formulas

- ▶ $\sin(90^\circ - \theta) = \cos \theta$ | $\cos(90^\circ - \theta) = \sin \theta$
- ▶ $\tan(90^\circ - \theta) = \cot \theta$ | $\cot(90^\circ - \theta) = \tan \theta$
- ▶ $\sec(90^\circ - \theta) = \operatorname{cosec} \theta$ | $\operatorname{cosec}(90^\circ - \theta) = \sec \theta$

⚠ Common Mistakes

- $\sin^2\theta + \cos^2\theta = 1$, NOT $\sin\theta + \cos\theta = 1$.
- $\tan 90^\circ$ and $\cot 0^\circ$ are UNDEFINED — never evaluate these.
- $\sin(A+B) \neq \sin A + \sin B$ (a very common algebraic error).
- Complementary: $\sin(90^\circ - \theta) = \cos\theta$. Always pair $\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \operatorname{cosec}$.

8.5 Worked Example

➡ Worked Example

Q: Prove: $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$

Step 1: Expand LHS: $\sin^2\theta + 2 \cdot \sin\theta \cdot \operatorname{cosec}\theta + \operatorname{cosec}^2\theta + \cos^2\theta + 2 \cdot \cos\theta \cdot \sec\theta + \sec^2\theta$

Step 2: $= \sin^2\theta + 2 + \operatorname{cosec}^2\theta + \cos^2\theta + 2 + \sec^2\theta$ (since $\sin\theta \cdot \operatorname{cosec}\theta = 1$, $\cos\theta \cdot \sec\theta = 1$)

Step 3: $= (\sin^2\theta + \cos^2\theta) + 4 + \operatorname{cosec}^2\theta + \sec^2\theta = 1 + 4 + (1 + \cot^2\theta) + (1 + \tan^2\theta)$

Step 4: $= 7 + \tan^2\theta + \cot^2\theta = \text{RHS}$

∴ **Answer: Proved ✓**

8.6 PYQ Triggers

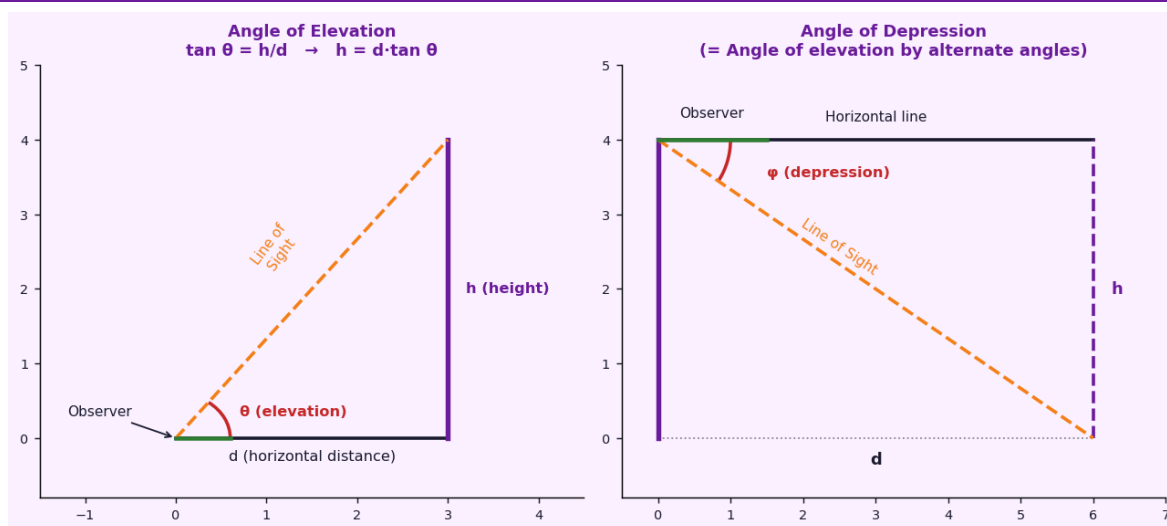
Year	PYQ — Board Exam Question
2024	If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = 1/\sqrt{3}$, find A and B. [2 marks]
2023	Prove: $(\cos A - \sin A + 1)/(\cos A + \sin A - 1) = \operatorname{cosec} A + \cot A$. [5 marks]
2022	Evaluate: $(\sin^2 63^\circ + \sin^2 27^\circ)/(\cos^2 17^\circ + \cos^2 73^\circ)$. [2 marks]
2021	If $\sec\theta + \tan\theta = p$, prove $\sin\theta = (p^2 - 1)/(p^2 + 1)$. [3 marks]
2019	Prove: $\cos A/(1 + \sin A) + (1 + \sin A)/\cos A = 2 \sec A$. [3 marks]

CHAPTER 9 · Unit V — Part of 12 Marks

Applications of Trigonometry

Heights & Distances | Angle of Elevation & Depression

9.1 Key Definitions & Diagram



Graph 9.1: Angle of elevation (left) and angle of depression (right)

Term	Definition	Direction
Line of sight	Line from observer's eye to the object	Towards the object
Angle of Elevation	Angle ABOVE horizontal (observer looks UP)	Measured upward from horizontal
Angle of Depression	Angle BELOW horizontal (observer looks DOWN)	Measured downward from horizontal

Critical Rules

- Angles in board exam problems: ONLY 30° , 45° , and 60° ($\tan 30^\circ = 1/\sqrt{3}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$)
- Angle of Elevation = Angle of Depression (alternate interior angles when observer and object are at same horizontal level)
- Always DRAW a neat diagram and label all known/unknown quantities FIRST

9.2 Core Formula

$$\tan \theta = \frac{\text{Height (h)}}{\text{Horizontal distance (d)}} \Rightarrow h = d \times \tan \theta$$

Problem-Solving Strategy (5 steps)

- Step 1: Read carefully — identify given and required.
- Step 2: Draw a CLEAR labelled diagram (right-angled triangle).
- Step 3: Mark the angle of elevation/depression and the sides.
- Step 4: Apply tan/sin/cos with the standard angle ($30^\circ/45^\circ/60^\circ$).
- Step 5: Solve algebraically; give the final answer with units.

⚠ Common Mistakes

- Elevation angle is measured from HORIZONTAL, not from vertical.
- Do NOT use sin or cos when tan is sufficient — $\tan = \text{height/distance}$ is most direct.
- When two angles from the SAME point: set up equations for the same triangle.
- Angle of depression viewed from a height = angle of elevation from the base (alternate angles).

9.3 Worked Example

➡ Worked Example

Q: A 1.5 m tall boy stands 28.5 m from a 30 m tower. Find angle of elevation of top.

Step 1: Height of tower above boy's eye level = $30 - 1.5 = 28.5$ m

Step 2: Horizontal distance = 28.5 m

Step 3: $\tan \theta = 28.5 / 28.5 = 1 \Rightarrow \theta = 45^\circ$

∴ Answer: Angle of elevation = 45°

9.4 PYQ Triggers

Year	PYQ — Board Exam Question
2024	From ground, angles of elevation of bottom and top of 20 m building on 60 m tower are 45° and 60° . Find distance of point. [4 marks]
2023	Two poles 6 m and 11 m stand 12 m apart. Find distance between their tops. [3 marks]
2022	Plane at 3000 m observes cars on opposite sides at depression angles 60° and 45° . Find distance between cars. [4 marks]
2021	Elevation changes 30° to 60° as observer moves 20 m towards a tower. Find height. [4 marks]
2019	A 1.2 m girl spots a ball at 60° elevation, 12 m away. Find height of ball. [2 marks]

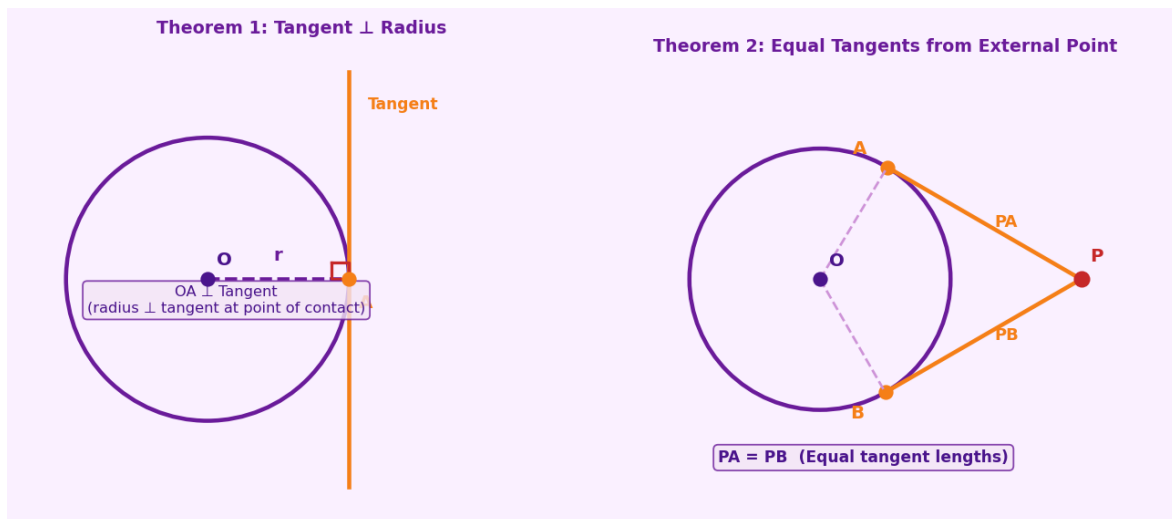
CHAPTER 10 · Unit IV — Part of 15 Marks

Circles

Tangent Properties | Equal Tangents Theorem | Angle Relationships

10.1 Key Terms & Number of Tangents

- ▶ Tangent: line touching the circle at exactly ONE point (point of tangency)
- ▶ Secant: line intersecting the circle at TWO points
- ▶ From a point OUTSIDE the circle: exactly 2 tangents can be drawn
- ▶ From a point ON the circle: exactly 1 tangent (the tangent at that point)
- ▶ From a point INSIDE the circle: NO tangent can be drawn (0 tangents)



Graph 10.1: Tangent $\perp$ radius (left) and equal tangents from external point (right)

10.2 Key Theorems (Both proofs required in exams)

Theorem 1: Tangent is Perpendicular to Radius at Point of Contact

- The tangent at any point of a circle is perpendicular to the radius at the point of contact.
- i.e. $OA \perp PA$, where O = centre, A = point of tangency, P = external point
- Proof method: assume not $\perp$, show a shorter line from centre exists — contradiction

Theorem 2: Lengths of Tangents from an External Point are Equal

- $PA = PB$, where P is the external point, A and B are the two points of tangency
- Proof: $\triangle OAP \cong \triangle OBP$ by RHS ($OA=OB=\text{radius}$, $OP=OP$, $\angle OAP=\angle OBP=90^\circ$) $\Rightarrow PA = PB$
- Also: $\angle OPA = \angle OPB$ (tangents equally inclined to line joining external point to centre)

- ▶ $PA = PB$ (equal tangent lengths from external point P)

► In right $\triangle OAP$ (Pythagoras): $OP^2 = OA^2 + AP^2$ $OA = \text{radius}, OA \perp AP$

► $\angle APB + \angle AOB = 180^\circ$ (angles at P and O in quadrilateral OAPB are supplementary)

⚠ Common Mistakes

- $PA = PB$ refers to LENGTHS, not angles.
- The right angle is at A (and B), NOT at P or O.
- RHS congruence proof: right angle at A, hypotenuse OP is common, $OA = OB = \text{radius}$.
- Do NOT confuse tangent-radius theorem with equal tangent theorem — both may appear.

10.3 Worked Example

➡ Worked Example

Q: Two tangents PA and PB are drawn from external point P. $\angle APB = 80^\circ$. Find $\angle AOB$.

Step 1: $\angle OAP = \angle OBP = 90^\circ$ (tangent $\perp$ radius)

Step 2: In quadrilateral OAPB: sum of angles = 360°

Step 3: $\angle AOB + 90^\circ + 80^\circ + 90^\circ = 360^\circ$

Step 4: $\angle AOB = 360^\circ - 260^\circ = 100^\circ$

$\therefore$ Answer: $\angle AOB = 100^\circ$

10.4 PYQ Triggers

Year	PYQ — Board Exam Question
2024	P is 10 cm from centre of a circle of radius 6 cm. Find the tangent length. [2 marks]
2023	Prove that tangents from an external point are equal in length. [4 marks]
2022	Two concentric circles, radii 5 and 13. Chord of larger is tangent to smaller. Find chord length. [3 marks]
2021	Tangent from P makes 60° with OP. Circle has radius r. Find OP. [2 marks]
2019	PQ is tangent from P; PA is secant. $PQ = 6$, $PB = 2$. Find PA. [2 marks]

CHAPTER 11 · Unit VI — Part of 10 Marks

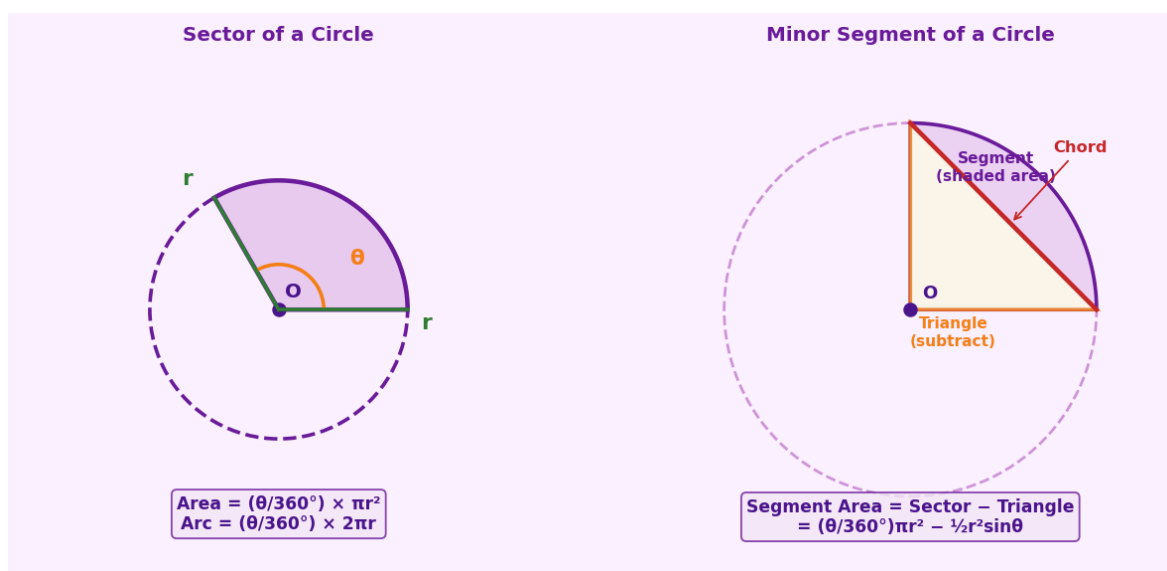
Areas Related to Circles

Circle | Sector | Segment | Combination of Figures

11.1 Standard Formulas

Figure	Area	Perimeter / Length
Circle (radius r)	πr^2	$2\pi r$
Semicircle	$\pi r^2/2$	$\pi r + 2r$
Quadrant (quarter)	$\pi r^2/4$	$\pi r/2 + 2r$
Sector (angle θ°)	$(\theta/360) \times \pi r^2$	$(\theta/360) \times 2\pi r$
Arc length (angle θ°)	—	$(\theta/360) \times 2\pi r$
Minor Segment	Area of sector – Area of Δ	—

- **Area of Sector** = $\frac{\theta}{360^\circ} \times \pi r^2$ *θ in degrees*
- **Length of Arc** = $\frac{\theta}{360^\circ} \times 2\pi r$
- **Area of Minor Segment** = $\frac{\theta}{360^\circ} \times \pi r^2 - \frac{1}{2}r^2 \sin\theta$



Graph 11.1: Sector of a circle (left) and minor segment (right) — Segment = Sector – Triangle

Key Values to Memorise

- $\pi = 22/7$ when radius has 7 as a factor | $\pi = 3.14$ otherwise
- For 90° (quadrant): Area = $\pi r^2/4$, Arc = $\pi r/2$
- For 120° sector: Area = $\pi r^2/3$, Arc = $2\pi r/3$

- For 60° sector: Area = $\pi r^2/6$, Arc = $\pi r/3$

⚠ Common Mistakes

- Segment $\neq$ Sector. Segment = Sector – Triangle (the chord cuts off the triangle).
- Perimeter of sector = $2 \times \text{radius} + \text{arc length}$ (NOT just arc length).
- Match π to the radius: use $22/7$ if r is a multiple of 7; otherwise use 3.14.

11.2 Worked Example

➡ Worked Example

Q: Find the area of a sector of radius 7 cm with central angle 90° . Use $\pi = 22/7$.

Step 1: Area = $(\theta/360) \times \pi r^2$

Step 2: = $(90/360) \times (22/7) \times 7^2$

Step 3: = $(1/4) \times (22/7) \times 49 = (1/4) \times 154$

∴ Answer: Area = 38.5 cm^2

11.3 PYQ Triggers

Year	PYQ — Board Exam Question
2024	Square of side 14 cm with quadrant of radius 7 cm at each corner cut out. Find remaining area. [3 marks]
2023	Radius of brooch = 21 mm. 5 equal sectors; one replaced by floral design. Find its area. [4 marks]
2022	Perimeter of a sector is 45 cm, radius 12 cm. Find its area. [3 marks]
2021	Horse tied at corner of $40 \text{ m} \times 40 \text{ m}$ field, rope = 14 m. Find ungrazed area. [4 marks]
2019	Rectangle $8 \times 20 \text{ cm}$. Two semicircles of radius 4 cut from shorter sides. Find shaded area. [4 marks]

CHAPTER 12 · Unit VI — Part of 10 Marks

Surface Areas and Volumes

All 3D Shapes | Combinations | Frustum of a Cone

12.1 Complete Formula Reference

Shape	Curved/Lateral SA	Total SA	Volume
Cube (a)	$4a^2$	$6a^2$	a^3
Cuboid (l,b,h)	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Cylinder (r,h)	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Cone (r,l,h)	πrl	$\pi r(r+l)$	$\pi r^2 h/3$
Sphere (r)	$4\pi r^2$	$4\pi r^2$	$4\pi r^3/3$
Hemisphere (r)	$2\pi r^2$	$3\pi r^2$	$2\pi r^3/3$
Frustum (R,r,l)	$\pi(R+r)l$	$\pi[(R+r)l+R^2+r^2]$	$\pi h(R^2+Rr+r^2)/3$

Cone Slant Height and Frustum Slant Height

► Cone slant height: $l = \sqrt{h^2 + r^2}$ ► Frustum slant height: $l = \sqrt{h^2 + (R - r)^2}$ *R = bigger base radius, r = smaller base radius*

Frequently Tested Combinations

- Cone on cylinder: CSA = $\pi rl + 2\pi rh$ (sum of curved surfaces of each)
- Hemisphere on cylinder: TSA = $2\pi r^2 + 2\pi rh + \pi r^2 = 3\pi r^2 + 2\pi rh$
- Ice-cream cone: Volume = Volume of cone + Volume of hemisphere
- Metal sphere melted into cylinders: total volume is CONSERVED

Common Mistakes

- Cone slant height $l = \sqrt{h^2 + r^2}$ — use this when l is not directly given.
- Frustum slant height: $\sqrt{h^2 + (R-r)^2}$, NOT $\sqrt{h^2 + R^2 - r^2}$.
- Combined shape TSA: do NOT count the HIDDEN face where two shapes join.
- Hemisphere TSA = $3\pi r^2$ (curved $2\pi r^2$ + flat base πr^2), NOT $4\pi r^2$.

12.2 Worked Example

— Worked Example

Q: A solid toy: cylinder with hemispherical ends. Total length = 2.7 m, diameter = 0.7 m. Find total SA.

Step 1: $r = 0.35$ m, cylinder height $h = 2.7 - 2(0.35) = 2.0$ m

Step 2: $TSA = CSA \text{ of cylinder} + 2 \times CSA \text{ of hemisphere}$

Step 3: $= 2\pi rh + 2 \times 2\pi r^2 = 2\pi(0.35)(2) + 4\pi(0.35)^2$

Step 4: $= 2\pi(0.70 + 0.49) = 2\pi \times 1.19 \approx 7.48 \text{ m}^2$

$\therefore$ Answer: $TSA \approx 7.48 \text{ m}^2$

12.3 PYQ Triggers

Year	PYQ — Board Exam Question
2024	Wooden toy: cylinder topped with cone. Cone height=2 cm, radius=3.5 cm, total height=15.5 cm. Find volume. [4 marks]
2023	Metallic sphere of radius 4.2 cm melted into small cylinders of radius 0.6 cm, height 1.4 cm. How many? [4 marks]
2022	Water flows through pipe of radius 1 cm at 80 cm/s. Time to fill tank $40 \times 20 \times 36$ cm. [4 marks]
2021	Frustum bucket: top radius 30 cm, bottom 10 cm, height 24 cm. Find slant height, CSA, metal sheet needed. [5 marks]
2019	Solid hemisphere on cylinder. Diameter = 7 cm, cylindrical height = 10 cm. Find TSA. ($\pi=22/7$) [4 marks]

CHAPTER 13 · Unit VII — Part of 11 Marks

Statistics

Mean | Median | Mode | Grouped Data

13.1 Mean — Three Methods for Grouped Data

Direct Method

$$\text{Mean } \bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Assumed Mean Method

► Choose a = assumed mean (usually midpoint of middle class)

$$\text{Mean} = a + \frac{\sum f_i d_i}{\sum f_i} \quad d_i = x_i - a$$

Step Deviation Method (easiest for large values)

$$\text{Mean} = a + \frac{\sum f_i u_i}{\sum f_i} \times h \quad u_i = (x_i - a)/h, \quad h = \text{class width}$$

13.2 Mode

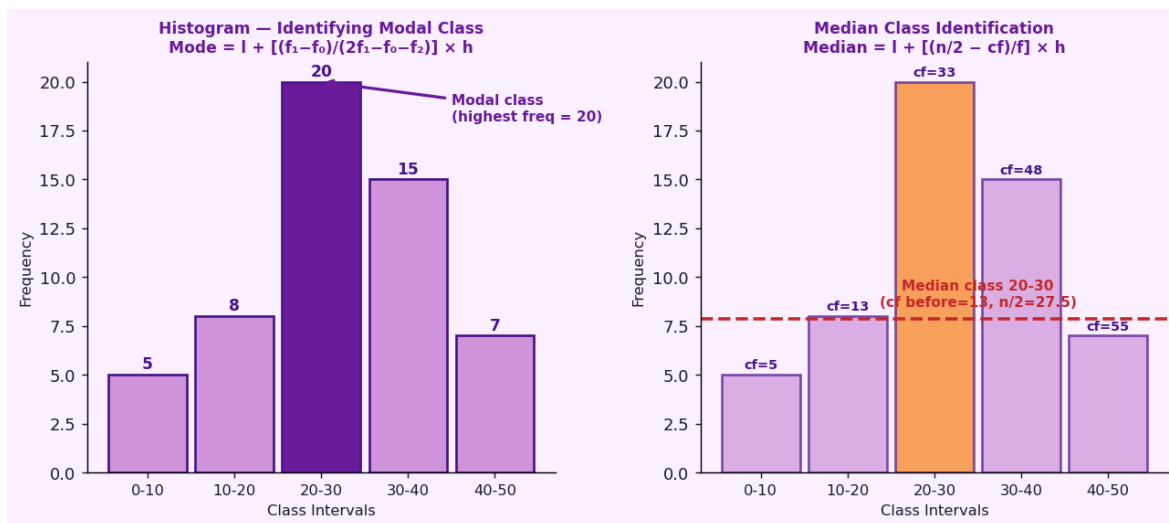
$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

Symbol	Meaning
l	Lower class boundary of modal class
f_1	Frequency of modal class (highest frequency)
f_0	Frequency of class BEFORE modal class
f_2	Frequency of class AFTER modal class
h	Class width (size of each interval)

13.3 Median

$$\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$$

Symbol	Meaning
l	Lower class boundary of median class
n	Total frequency ($\sum f$)
cf	Cumulative frequency of class BEFORE the median class
f	Frequency of the median class
h	Class width



Graph 13.1: Histogram with modal class highlighted (left) and median class identification (right)

Key Results

- Median class: first class where cumulative frequency $\geq n/2$
- Modal class: class with the HIGHEST frequency
- Empirical relation: $\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$

Common Mistakes

- Mode formula denominator: $2f_1 - f_0 - f_2$ (students often write only $f_1 - f_0$).
- Median: cf is cumulative frequency BEFORE (not including) the median class.
- Both l in mode and median = LOWER class boundary (not midpoint).

13.4 Worked Example

Worked Example

Q: Find the mean: classes 0–20, 20–40, 40–60, 60–80 with frequencies 6, 8, 10, 6.

Step 1: x_i (midpoints) = 10, 30, 50, 70

Step 2: $f_i x_i$ = 60, 240, 500, 420

Step 3: $\sum f_i = 30$ | $\sum f_i x_i = 1220$

Step 4: Mean = $1220 / 30$

∴ Answer: Mean ≈ 40.67

13.5 PYQ Triggers

Year	PYQ — Board Exam Question
2024	Find missing frequencies if mean = 50: classes 0-20, 20-40, 40-60, 60-80, 80-100 with freq 17, f_1 , 32, f_2 , 19; total=120. [4 marks]

2023	Find median: 0-10(5), 10-20(8), 20-30(20), 30-40(15), 40-50(7), 50-60(5). [3 marks]
2022	Daily wages: 100-120(10), 120-140(15), 140-160(20), 160-180(22), 180-200(18). Find mode and median. [5 marks]
2021	Find mean of the following data using Step Deviation method: 0-10(5), 10-20(8), 20-30(20), 30-40(15), 40-50(7). [3 marks]
2019	Find mode for 25-35(7), 35-45(31), 45-55(33), 55-65(17), 65-75(11), 75-85(1). [2 marks]

CHAPTER 14 · Unit VII — Part of 11 Marks

Probability

Classical Definition | Events | Complementary Probability | Cards & Dice

14.1 Basic Definitions & Formula

- ▶ Experiment: an action with well-defined outcomes
 - ▶ Sample Space (S): the set of ALL possible outcomes of an experiment
 - ▶ Event (E): any subset of the sample space
 - ▶ Favourable outcomes: the outcomes that satisfy the event E
- ▶ $P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Total number of equally likely outcomes}}$ *Classical (Theoretical) definition*
- ▶ $0 \leq P(E) \leq 1$ for any event E
 - ▶ $P(E) + P(\bar{E}) = 1 \Rightarrow P(\bar{E}) = 1 - P(E)$ (complementary events)
 - ▶ $P(\text{impossible event}) = 0 \quad | \quad P(\text{certain event}) = 1$

14.2 Standard Sample Spaces

Experiment	Sample Space	n(S)
Toss 1 coin	H, T	2
Toss 2 coins	HH, HT, TH, TT	4
Toss 3 coins	HHH, HHT, HTH, HTT, ...	8
Roll 1 die	1, 2, 3, 4, 5, 6	6
Roll 2 dice	(1,1), (1,2), ..., (6,6)	36
Pick 1 card from deck	All 52 cards	52
Pick from red cards	26 red cards (Hearts + Diamonds)	26

Card Deck — Essential Facts for Exams

- 52 cards total: 4 suits (Spades ♠, Hearts ♥, Diamonds ♦, Clubs ♣), 13 cards per suit
- Face cards: Jack, Queen, King = 3 per suit = 12 face cards total
- Aces: 4 total (one per suit)
- Red cards: Hearts + Diamonds = 26 | Black cards: Spades + Clubs = 26
- Number cards: 2 through 10 = 9 per suit = 36 number cards total

Use the Complement Rule (saves time!)

- $P(\text{at least one head}) = 1 - P(\text{no head})$ — always faster than listing all cases
- $P(\text{not a multiple of 3}) = 1 - P(\text{multiple of 3})$
- $P(\text{at least one}) = 1 - P(\text{none})$ (general rule)

⚠ Common Mistakes

- P = favourable / total, NOT total / favourable.
- Face cards do NOT include Aces. Face cards = J, Q, K only (12 cards).
- Ace of spades is a black Ace; Ace of hearts is a red Ace.
- For two dice: sample space has 36 outcomes, NOT 12.

14.3 Worked Example

➡ Worked Example

Q: Two dice are thrown simultaneously. Find $P(\text{sum} = 7)$.

Step 1: Total outcomes = $6 \times 6 = 36$

Step 2: Favourable outcomes (sum=7): (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) = 6 outcomes

Step 3: $P(\text{sum} = 7) = 6/36$

∴ Answer: $P(\text{sum} = 7) = 1/6$

14.4 PYQ Triggers

Year	PYQ — Board Exam Question
2024	Bag: 6 red, 4 white, 8 blue balls. One drawn at random. Find $P(\text{not blue})$. [1 mark]
2023	Cards 1–25 in a box. $P(\text{number divisible by 2 or 3})$. [2 marks]
2022	Die thrown twice. $P(\text{first throw} < 6 \text{ and product} < 8)$. [2 marks]
2021	From 52 cards: find $P(\text{king of red suit})$ and $P(\text{face card})$. [2 marks]
2019	Number x chosen from 1–15. Find $P(x^2 < 0) \mid P(2x+4 > 14) \mid P(3x+1 = 8)$. [2 marks]

Master Quick Revision Sheet — All 14 Chapters (2025-26)

Chapter	★ Must-Know Formulas & Key Points
Ch 1 Real Numbers	$HCF \times LCM = a \times b$ Proof: assume p/q , $HCF(p,q)=1$, derive contradiction
Ch 2 Polynomials	$\alpha + \beta = -b/a$ $\alpha\beta = c/a$ $poly = x^2 - (\alpha + \beta)x + \alpha\beta$
Ch 3 Linear Equations	$a_1/a_2 \neq b_1/b_2$ unique $=c$ ratio $\rightarrow \infty$ $=b \neq c$ ratio $\rightarrow$ no soln
Ch 4 Quadratic Equations	$x = (-b \pm \sqrt{D})/2a$ $D > 0$: 2 roots, $D = 0$: equal, $D < 0$: no real roots
Ch 5 AP	$a_n = a + (n-1)d$ $S_n = n/2[2a + (n-1)d]$ 3 terms: $(a-d)$, a , $(a+d)$
Ch 6 Triangles	BPT: $AD/DB = AE/EC$ Area ratio = side ratio ² $c^2 = a^2 + b^2$
Ch 7 Coord. Geometry	$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ Section: $(mx_2 + nx_1)/(m+n)$
Ch 8 Trigonometry	$\sin^2\theta + \cos^2\theta = 1$ SOH CAH TOA $\sin(90^\circ - \theta) = \cos\theta$
Ch 9 Heights & Dist.	$\tan\theta = h/d$ angles only $30^\circ/45^\circ/60^\circ$ DRAW diagram always
Ch 10 Circles	$PA = PB$ (equal tangents) $OA \perp PA$ (tangent $\perp$ radius) $\angle AOB + \angle APB = 180^\circ$
Ch 11 Areas $\rightarrow$ Circles	Sector = $(\theta/360)\pi r^2$ Arc = $(\theta/360)2\pi r$ Segment = Sector - Δ
Ch 12 SA & Volumes	Cone: $V = \pi r^2 h/3$, $l = \sqrt{h^2 + r^2}$ Sphere: $V = 4\pi r^3/3$ Frustum $l = \sqrt{h^2 + (R-r)^2}$
Ch 13 Statistics	Mode = $l + [(f_1 - f_0)/(2f_1 - f_0 - f_2)] \times h$ Median: find $cf \geq n/2$ Mode $\approx 3\text{Med} - 2\text{Mean}$
Ch 14 Probability	$P(E) = \text{favourable}/\text{total}$ $P(\bar{E}) = 1 - P(E)$ 52 cards: 4 suits, 13 each

Board Exam Strategy 2025-26 (80 Marks, 3 Hours)

Strategy	Detail
Paper Pattern	20 MCQ (1M) + 5 VSA (2M) + 6 SA (3M) + 4 LA (5M) + Case Study (4M)
Time Allocation	MCQ: 20 min 2M: 2.5 min each 3M: 5 min 5M: 8 min Case study: 10 min
Step Marking	Write every step. Stating the formula earns marks even if final answer is wrong.
Proof Questions	State theorem $\rightarrow$ write Given, To Prove, Construction, Proof with justifications.
Geometry Diagrams	Draw neat, large diagrams; label all points, sides, angles and measurements.
High Yield Units	Algebra (20M) $\rightarrow$ Triangles+Circles (15M) $\rightarrow$ Trig (12M) $\rightarrow$ Stats+Prob (11M)
Internal Choices	Pick the question whose concept you know better; don't attempt both.
Case-Based Qs	Read all sub-questions first, then re-read the passage with context.
Key Formulas	Memorise: quadratic formula, sector area, AP nth term & sum, trig values table.
Revision Schedule	This book: 1 week before $\rightarrow$ 3 days before $\rightarrow$ morning of exam.

Final Words from SJ Maths

- This book covers 100% of the official CBSE 2025-26 Class 10 Maths syllabus.
- For each chapter: understand the concept → study the graphs → apply the formulas → practice PYQs.
- Visit sjmaths.com for free video solutions to every PYQ listed in this book.
- Consistency beats cramming. 30 minutes daily > 3 hours the night before.
- All the very best for your CBSE Board Exams 2025-26! 🍀